

Introduction

Social isolation in the digital age has been an increasingly concerning topic. First it posed a solution to social anxiety and isolation, then it quickly became the concern for causing these issues. As some researchers look for various ways to mitigate these social concerns through limiting use of mainstream technology, others are looking for new ways to use these technologies as a tool. Virtual Reality (VR) is one of these emerging technologies piquing the interest of researchers across various areas of study. Some social scientists research the negative implications of VR and what sorts of impacts it may have on the psyche. Others research ways these emerging immersive technologies can be used to improve already-existing issues. This literature review will focus on the latter of the two groups, examining empirical studies related to the efficacy of VR in improving social ability and wellness.

In-Depth Summary

One pilot study done by Rus-Calafell et.al (2014) was conducted in a VR program for improving social skills in patients with Schizophrenia. A program was set up in VR so therapists could run instances with the participants to practice social interactions in a structured, role-play-like setting. Qualitative analyses and quantitative surveys were taken at the end of the program and at the post-test check-in. Participants showed improvement in social functioning and decrease in negative symptoms, but in the follow-up check-in 4 months later their discomfort and anxiety scales had increased again- likely due to lack of maintenance.

This study is important to note because it shows that even people who struggle heavily with social interaction and mental health can use VR as a means of social wellness and improvement. This study employed a pre and post test design to properly capture the differences in how each subject felt after spending time in VR- a method I would like to employ myself. Scales- such as the Positive and Negative Syndrome Scale (PANSS)- were used to measure observed reactions and the patients' feelings in the study. I'd like to consider using this study's operationalization and research design in my own study.

Common Research Methods

Many studies have employed similar methodologies to analyze VR and social wellbeing. Utilizing environments in virtual reality to control and measure interactions during these studies is one very common design used by researchers in social experiments (Ke et al. 2020; Kenyon et al. 2023; Rus-Calafell et al. 2014). Building role-play-like environments to simulate real-world interactions in a virtual reality environment was one repeated approach (Ke et al. 2020; Rus-Calafell et al. 2014). Another similar approach was to use preexisting social spaces in VR, such as Altspace VR to conduct the study (Kenyon et al. 2023). It's common to find mixed-methodologies in social science VR surveys, particularly these ones related to social skills and wellbeing.

Multiple studies used pre and post test designs to measure effectiveness, with user performance and feelings measured before and after testing (Ayobi et al. 2023; Ke et al. 2020; Rus-Calafell et al. 2014). Another common method, taking a more qualitative approach, was to incorporate various surveys to assess feelings, behaviors and skill level

either before or after being in VR (Anderson et al. 2003; Ayobi et al. 2023; Ke et al. 2020; Kenyon et al. 2023; Rus-Calafell et al. 2014). Several studies conducted these behavioral assessments in tandem with observation to gauge physical cues- like eye contact and engagement- to evaluate participants (Anderson et al. 2003; Ke et al. 2020; Rus-Calafell et al. 2014). Overall, research tends to use a mixed-method approach, mixing qualitative interview and assessment with quantitative survey data and measurements. These have allowed researchers to explore experiences and include actual participants feelings in the evaluations, while having numerical values to assess the overall outcomes.

Definitions

Social skills, as defined by the American Psychological Association, are “a set of learned abilities that enable an individual to interact competently and appropriately in a given social context.” Social skills can be defined broadly as the ability to be competently able to respond and understand cues in given social situations and interactions. Virtual Reality, in my words, is a simulated, digital environment that can alter the user’s perception through immersion, translated through motion tracking technology and digital screens. The motion tracking technology is what sets it apart from all other modern technology; without it, it would be little more than a computer screen for your face.

Common Findings

Research commonly found that, after being immersed in the headset in either simulated or real interaction, social interaction ability was improved (Key et al. 2020; Kenyon et. al 2023; Rus-Calafell et al. 2014). It was also found that by participating in social

VR spaces people felt less isolated, lonely, or felt a reduced sense of anxiety around socializing (Anderson et al. 2003; Ayobi et. al. 2023). It's also important to note that some studies did note that VR may not replicate real-world interactions well enough or frequently enough to have a lasting effect (Anderson et al. 2003; Rus-Calafell et al. 2014). Some researchers reported participants having a hard time transferring their social skills from VR to real-life interactions (Ke et al. 2020; Rus-Calafell et al. 2014).

Summary and Critique

Virtual reality, though digital, can foster very real connections and interactions through psychologically changing the user's sense of presence, and presenting them with real scenarios and interactions to learn from. However, though it can be used as a tool to increase social ability and wellbeing, it also has limitations. Its long-term efficacy and real world application may be limited if there isn't applied practice outside of the virtual reality. Future research should explore a hybrid approach to integrate virtual reality with the real world, implementing VR first and the real world later. Additionally, longitudinal studies should be performed to assess the long-term impact of VR on mental health and social functioning, and how it might be turned to as an escape from reality rather than a tool to enhance it.

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